Non-negativity of the trace core for the staircase Bott–Samelson operator on Hecke cell modules

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Abstract

Let $H_q = H_q(\mathfrak{S}_n)$ be the Hecke algebra of the symmetric group with parameter q, and let $\Pi_q = \prod_{j=1}^{\binom{n}{2}}(1+T_{i_j})$ be the Bott–Samelson product along the staircase reduced word $\underline{w}_0 = s_1 \cdot s_2 s_1 \cdot s_3 s_2 s_1 \cdots s_{n-1} \cdots s_1$ for the longest element. For any partition $\lambda \vdash n$ let V_λ denote the corresponding cell module. Then $\sigma_1(V_\lambda) := \mathrm{tr}(\Pi_q|_{V_\lambda}) \in \mathbb{Z}[q]$ is palindromic (Apr 23 theorem) and non-negative (by Kazhdan–Lusztig positivity). We prove:

If $\sigma_1(V_\lambda) = q^{a_\lambda}(1+q)^{b_\lambda}P_\lambda(q)$ with (a_λ, b_λ) maximal, then $P_\lambda(q) \in \mathbb{Z}_{\geq 0}[q]$.

The proof gives an explicit decomposition $\sigma_1 = \sum_k n_k(\lambda)(1+q)^k q^{(N-k)/2}$ with $n_k(\lambda) \in \mathbb{Z}_{\geq 0}$, summed over $k \equiv N \pmod 2$, where $N = \binom{n}{2}$. Verified on all λ with $V_\lambda^H \neq 0$ at $n \leq 7$.

1 Setup and main theorem

Let H_q be the Hecke algebra of $\mathfrak{S}_n$ over $\mathbb{Z}[q, q^{-1}]$, with generators $T_1, \ldots, T_{n-1}$ satisfying

$$T_i^2 = (q-1)T_i + q, \qquad T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \qquad T_iT_j = T_jT_i \ (|i-j| \geq 2).$$

Working in $H_q \otimes_{\mathbb{Z}[q]} \mathbb{Z}[v, v^{-1}]$ with $q = v^2$ allows us to use the Kazhdan–Lusztig basis. For each partition $\lambda \vdash n$ we have the cell (Specht) module V_λ; it carries a canonical Kazhdan–Lusztig basis $\{b_w : w \in \mathcal{C}_\lambda\}$ indexed by a left cell $\mathcal{C}_\lambda$ in $\mathfrak{S}_n$, and equivalently by standard Young tableaux of shape λ. This basis is bar-invariant and exhibits the *W-graph positivity* of Kazhdan–Lusztig (in type A proved elementarily; in general by Elias–Williamson).

Fix the staircase reduced word for the longest element $w_0 \in \mathfrak{S}_n$:

$$\underline{w}_0 = (i_1, i_2, \ldots, i_N), \qquad N = \binom{n}{2},$$

where the sequence is $1;\ 2,1;\ 3,2,1;\ \ldots;\ n-1, n-2, \ldots, 1$. Define the *staircase Bott–Samelson operator*

$$\Pi_q := \prod_{j=1}^{N}(T_{i_j} + 1) \in H_q.$$

Theorem 1 (σ_1-G1)**.** *Let $\lambda \vdash n$ and $\sigma_1(V_\lambda) := \mathrm{tr}(\Pi_q|_{V_\lambda}) \in \mathbb{Z}[q]$. Then there exist non-negative integers $n_k(\lambda) \in \mathbb{Z}_{\geq 0}$, for k ranging over $\{0, 1, \ldots, N\}$ with $k \equiv N \pmod 2$, such that*

$$\sigma_1(V_\lambda) = \sum_{\substack{0 \leq k \leq N \\ k \equiv N \ (\mathrm{mod}\ 2)}} n_k(\lambda) \cdot (1+q)^k \cdot q^{(N-k)/2}. \tag{1}$$

Consequently, if $\sigma_1(V_\lambda) = q^{a_\lambda}(1+q)^{b_\lambda}P_\lambda(q)$ with (a_λ, b_λ) maximal, then

$$P_\lambda(q) = \sum_k n_k(\lambda) \cdot (1+q)^{k-b_\lambda} \cdot q^{(N-k)/2-a_\lambda} \in \mathbb{Z}_{\geq 0}[q].$$

2 The key decomposition of $T_i + 1$ in the KL basis

Lemma 2 (KL decomposition of $T_i + 1$)**.** *In the Kazhdan–Lusztig basis $\{b_w\}$ of V_λ, the action of $T_i + 1$ takes the form*

$$T_i + 1 = (1+q)\, D_i + v\, M_i,$$

where D_i and M_i are matrices with non-negative integer entries, satisfying:

[(i)]

1. *D_i is diagonal: $(D_i)_{ww} = [s_i \cdot w < w]$ (indicator that s_i is a left descent of w).*
2. *M_i is off-diagonal with entries in $\mathbb{Z}_{\geq 0}$, and $(M_i)_{z,w} \neq 0$ implies $\ell(z) - \ell(w)$ is* odd.

Proof. Recall the Kazhdan–Lusztig element $C'_{s_i} \in H_q$ acts on V_λ (in the KL basis) by

$$C'_{s_i} \cdot b_w = \begin{cases} (v+v^{-1})b_w & \text{if } s_i \cdot w < w, \\ b_{s_i \cdot w} + \displaystyle\sum_{\substack{z\,:\, s_i \cdot z < z \\ \mu(z,w) \neq 0}} \mu(z,w) b_z & \text{if } s_i \cdot w > w, \end{cases} \tag{2}$$

where $\mu(z,w)$ are the Kazhdan–Lusztig μ-coefficients (non-negative integers, by KL positivity). This is the defining action of C'_{s_i} on a W-graph cell module.

The relation $T_i + 1 = v\, C'_{s_i}$ (standard, with $q = v^2$ and the Hecke convention $T_i^2 = (q-1)T_i + q$) gives:

- If $s_i \cdot w < w$: $(T_i + 1)b_w = v(v + v^{-1})b_w = (v^2 + 1)b_w = (1+q)b_w$.
- If $s_i \cdot w > w$: $(T_i + 1)b_w = v \cdot b_{s_i w} + v \sum_z \mu(z,w) b_z$.

Separating the diagonal and off-diagonal parts yields the decomposition $(T_i + 1) = (1+q)D_i + vM_i$ with the stated properties. The entries of D_i are in $\{0,1\}$ and those of M_i are non-negative integers by KL positivity.

The parity statement in (ii) is the *parity vanishing* of KL polynomials for $\mathfrak{S}_n$: $\mu(z,w) \neq 0$ implies $\ell(w) - \ell(z)$ is odd (see [**?**, §6], or [**?**] for the general Soergel proof). In the case $z = s_i \cdot w$ with $s_i w > w$, we have $\ell(z) = \ell(w) + 1$, clearly odd. □

3 Expanding Π_q and computing the trace

Applying Lemma **??** at each factor:

$$\Pi_q = \prod_{j=1}^{N} \big[(1+q)D_{i_j} + vM_{i_j}\big] = \sum_{S \subseteq \{1,\ldots,N\}} (1+q)^{|S|} \cdot v^{N-|S|} \cdot \prod_{j=1}^{N} X_j^S, \tag{3}$$

where the outer product respects the order $j = 1, 2, \ldots, N$ and $X_j^S := D_{i_j}$ if $j \in S$, $X_j^S := M_{i_j}$ if $j \notin S$.

Lemma 3 (Parity of trace)**.** *For any $S \subseteq \{1, \dots, N\}$, $\operatorname{tr}\big(\prod_{j=1}^{N} X_j^S\big) \in \mathbb{Z}_{\geq 0}$, and the trace vanishes unless $|S^c| = N - |S|$ is even.*

Proof. Each D_i and M_i is a matrix with non-negative integer entries. Hence $\prod_j X_j^S$ has non-negative integer entries, and its trace is in $\mathbb{Z}_{\geq 0}$.

For the parity claim: the D-operators are diagonal and preserve length-parity. The M-operators, by Lemma **??**(ii), connect vertices of *opposite* length-parity. The diagonal entry of $\prod_j X_j^S$ at vertex w is a sum over closed paths $w = w_0 \to w_1 \to \cdots \to w_{N-1} \to w_N = w$ where, at step j, if $j \in S$ the path stays put ($w_j = w_{j-1}$) and the weight is $(D_{i_j})_{w_{j-1}, w_{j-1}}$; if $j \notin S$ the path may move and the weight is $(M_{i_j})_{w_{j-1}, w_j}$. Each M-step flips length-parity of the vertex, each D-step preserves it. For the path to close (return to w), the total number of parity flips must be even, i.e., $|S^c| = N - |S|$ is even. □

Proof of Theorem **??**. From (**??**) and Lemma **??**,

$$\sigma_1(V_\lambda) = \operatorname{tr}(\Pi_q|_{V\lambda}) = \sum_{\substack{S \subseteq \{1,\dots,N\} \\ |S^c| \text{ even}}} (1+q)^{|S|} v^{N-|S|} N_S,$$

where $N_S := \operatorname{tr}\big(\prod_j X_j^S\big) \in \mathbb{Z}_{\geq 0}$. Since $|S^c| = N - |S|$ is even, $v^{N-|S|} = q^{(N-|S|)/2}$ is a non-negative integer power of q. Grouping by $k := |S|$:

$$\sigma_1(V_\lambda) = \sum_{\substack{0 \leq k \leq N \\ k \equiv N \pmod 2}} (1+q)^k \, q^{(N-k)/2} \cdot \underbrace{\sum_{|S|=k} N_S}_{=: n_k(\lambda) \in \mathbb{Z}_{\geq 0}} .$$

This is equation (**??**).

For the non-negativity of P_λ: the $(1+q)$-adic valuation of σ_1 is $b_\lambda = \min\{k : n_k(\lambda) > 0\}$, and the q-adic valuation is $a_\lambda = \min_k\{(N-k)/2 : n_k(\lambda) > 0\} = (N - k_{\max})/2$ where $k_{\max} = \max\{k : n_k(\lambda) > 0\}$. (These equalities use that the terms $(1+q)^k q^{(N-k)/2}$ are coprime to each other with respect to both q and $1+q$, and that the n_k are non-negative so no cancellation occurs.)

Dividing both sides of (**??**) by $q^{a_\lambda}(1+q)^{b_\lambda}$:

$$P_\lambda(q) = \sum_k n_k(\lambda) \, (1+q)^{k-b_\lambda} \, q^{(N-k)/2 - a_\lambda}.$$

For every k with $n_k(\lambda) > 0$: $k \geq b_\lambda$ (so $(1+q)^{k-b_\lambda}$ is a non-negative polynomial) and $(N-k)/2 \geq a_\lambda$ (so $q^{(N-k)/2-a_\lambda}$ is a non-negative polynomial). Each summand is a non-negative polynomial in q, hence so is P_λ. □

4 Verification and example

Example 4 ($\lambda = (3,1)$ at $n = 4$)**.** Here $N = 6$ and $\sigma_1 = q(1+q)^2(q^2+4q+1) = q + 6q^2 + 10q^3 + 6q^4 + q^5$. Parity $N \equiv 0 \pmod 2$ forces k even. Solving the linear system (**??**) gives

$$n_0 = 0, \quad n_2 = 2, \quad n_4 = 1, \quad n_6 = 0,$$

so $\sigma_1 = 2\,q^2(1+q)^2 + q(1+q)^4$, each term manifestly in $\mathbb{Z}_{\geq 0}[q]$. Consequently $b_\lambda = 2$, $a_\lambda = 1$, and

$$P_\lambda = 2q + (1+q)^2 = q^2 + 4q + 1 \in \mathbb{Z}_{\geq 0}[q]. \quad \checkmark$$

Example 5 ($\lambda = (3,2,1)$ at $n = 6$)**.** $N = 15$ forces k odd. Solving gives

$$n_1 = 2, \quad n_3 = 15, \quad n_5 = 9, \quad n_7 = 1,$$

all other $n_k = 0$. So $b_\lambda = 1$, $a_\lambda = (15-7)/2 = 4$, and

$$P_\lambda = 2q^3 + 15q^2(1+q)^2 + 9q(1+q)^4 + (1+q)^6 = q^6 + 15q^5 + 66q^4 + 106q^3 + 66q^2 + 15q + 1. \quad \checkmark$$

Example 6 (Degree-10 novel orphan at $n = 7$)**.** For $\lambda = (3,2,1,1) \vdash 7$ at $k = 2$, one of the atoms of the factored σ_2 core is the irreducible degree-10 palindrome $1 + 16q + 105q^2 + 369q^3 + 759q^4 + 961q^5 + \cdots$ with $P(1) = 3461$ prime (Apr 24 work). Theorem **??** gives a non-negativity proof of P_λ for σ_1 only, so this example is *not* covered here; it remains open whether the analog σ_k-G1 holds. The novel orphan lives at $k = 2$, outside the scope of the present theorem.

Remark 7 (What is the scope?)*.* Theorem **??** covers σ_1 for *all* λ, both rank-1 and higher rank. We do *not* handle σ_k for $k \geq 2$. The key ingredient that fails for $k \geq 2$: the exterior power $\Lambda^k(\Pi_q)$ acting on $\Lambda^k V_\lambda$ does not admit the clean $D + vM$ decomposition. An alternative argument (e.g., via suitable Soergel multi-bimodules or tensor-power extensions) is required.

Remark 8 (What is used, what is needed)*.* The proof uses:

> [(a)]The existence of the Kazhdan–Lusztig basis of V_λ with the W-graph action (**??**). This exists over $\mathbb{Z}[v, v^{-1}]$ in type A. W-graph positivity: the μ-coefficients are non-negative integers. For $\mathfrak{S}_n$ this is elementary; in general (Elias–Williamson). Kazhdan–Lusztig parity vanishing: $\mu(z,w) \neq 0 \Rightarrow \ell(w) - \ell(z)$ odd.

No semisimplicity or genericity of q is required. The theorem holds with the same proof over any ring where the Hecke algebra and cell module are defined and the above positivity statements hold.

5 Computational verification

The decomposition (**??**) was verified computationally for all $\lambda \vdash n$ with $V_\lambda^H \neq 0$ at $n \leq 7$:

- $n = 3$: $\lambda \in \{(3), (2,1)\}$. All $n_k \geq 0$.
- $n = 4$: $\lambda \in \{(4), (3,1), (2,2)\}$. All $n_k \geq 0$.
- $n = 5$: $\lambda \in \{(5), (4,1), (3,2), (3,1,1), (2,2,1)\}$. All $n_k \geq 0$.
- $n = 6$: $\lambda \in \{(6), (5,1), (4,2), (4,1,1), (3,3), (3,2,1), (2,2,2)\}$. All $n_k \geq 0$.
- $n = 7$: $\lambda \in \{(7), (6,1), (5,2), (5,1,1), (4,3), (4,2,1), (4,1,1,1), (3,3,1), (3,2,2), (3,2,1,1), (2,2,2,1)\}$. All $n_k \geq 0$.

For λ with $V_\lambda^H = 0$ (e.g., $\lambda = (2, 1^{n-2})$ at $n \geq 4$), $\sigma_1 = 0$ trivially and all $n_k = 0$.

Scripts: `~/projects/scratch/2026-04-24-sigma1-G1/{compute_Pi_integral.py, verify_proof.py}`.

6 Discussion

What this solves

This resolves the G1 question for σ_1 (the trace of Π_q on each irreducible V_λ). Prior to today, the known properties of σ_1 were:

- $\sigma_1 \in \mathbb{Z}[q]$ (integrality, from cell-module integral form).
- σ_1 is palindromic of degree N (from bar-involution, Apr 23).
- $\sigma_1 \in \mathbb{Z}_{\geq 0}[q]$ (from KL positivity, by taking trace of a non-negative matrix).
- $\sigma_1(q) > 0$ for all $q > 0$ (eigenvalue positivity, proved Apr 16).

These do *not* force the non-negativity of the core P_λ after extracting $q^{a_\lambda}(1+q)^{b_\lambda}$: the palindromic cyclotomic $\Phi_6(q) = q^2 - q + 1$ satisfies all four properties yet has a negative coefficient. So G1 requires genuinely new structure.

The new structure is the decomposition (**??**): the $(1+q)$-factors and q-factors appearing in σ_1 are *not* artifacts of extraction; they are built in, one factor of $(1+q)$ per "D-step" (descent position) and one factor of q per "two M-steps" (KL-graph traversal).

What this does not solve

This does not address:

- σ_k-G1 for $k \geq 2$ (exterior power cores). The novel degree-10 orphan palindrome at $n = 7$, $\lambda = (3, 2, 1, 1)$, $k = 2$, is *not* covered.
- The atom alphabet question: are all P_λ (for σ_1) built from a finite set of irreducible palindromic atoms as n grows?
- Identification of b_λ and a_λ with combinatorial invariants of λ. The empirical rule "$b_\lambda = \lfloor n/2 \rfloor$ in the rank-1 case" remains conjectural; here it arises as $\min\{k : n_k(\lambda) > 0\}$ which the proof does not compute explicitly.

Extension to higher σ_k

For $\sigma_k(V_\lambda) = \mathrm{tr}(\Lambda^k \Pi_q|_{\Lambda^k V_\lambda})$, the natural generalization would be: does $\Lambda^k \Pi_q$ admit a decomposition analogous to Lemma **??**, with $(1+q)^{\text{descents}}$ and v^{edges} contributions? The exterior power of the individual factors $(T_i + 1) = (1+q)D_i + vM_i$ is

$$\Lambda^k(T_i + 1) = \sum_{j=0}^{k} (1+q)^j v^{k-j} \Lambda^j D_i \otimes \Lambda^{k-j} M_i,$$

but $\Lambda^{k-j} M_i$ need not have non-negative entries in the induced basis (a minor of a non-negative matrix can be negative). This appears to be the precise obstruction to extending the present proof.

References

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